LLM-Based Volunteer Coordination for Crowd Evacuation

A Comparison with a Game-Theoretic Model

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Abstract

Volunteer coordination can affect evacuation efficiency when obstacles obstruct access to exits. This study investigates an approach that uses a large language model (LLM) to support volunteer decisions within a grid-based crowd evacuation simulator. The approach replaces the game-theoretic volunteer decision module while retaining the surrounding movement, environment, and simulation framework. Evacuation context, including nearby pedestrians, visible obstacles, candidate volunteers, distances, and estimated obstacle impact, is used to inform decisions. Experiments cover four agent populations (60, 90, 120, and 150), with 30 runs per scenario. Performance is evaluated by T90, the number of simulation steps required for 90% of pedestrians to evacuate. The reported plots show lower T90 values for the LLM-based approach at all evaluated populations, with a larger difference at higher populations. Natural-language explanations provide an additional record of the contextual factors associated with volunteer decisions. These findings indicate potential benefits within the evaluated simulation setting; statistical significance and operational performance require further evaluation.

Keywords: crowd evacuation; volunteer coordination; large language models; cellular automata; game theory.

1 Introduction

Crowd evacuation simulation supports the study of pedestrian movement during emergencies such as fires and earthquakes. Cellular automata (CA) and floor-field methods provide a discrete representation of pedestrian movement toward exits. When obstacles block evacuation routes, movement alone may not restore access; pedestrians may also need to volunteer to remove obstructions.

The framework developed by Lin and Wong [1] incorporates obstacle removal and game-theoretic pedestrian behavior. Building on this setting, the present study examines whether contextual LLM reasoning can improve volunteer coordination relative to a game-theoretic decision model. The comparison focuses on the volunteer decision module while retaining the other simulation components.

The contributions are an LLM-based volunteer decision framework, a comparison with the game-theoretic baseline under a shared simulation design, and an examination of both evacuation performance and natural-language decision explanations.

2 Background and Baseline Method

Lin and Wong [1] integrate cellular automata and game theory into an evacuation model that considers obstacle removal. Their framework includes mechanisms for pedestrian conflict resolution and volunteering, together with an anticipation floor field. This study uses the obstacle-removal setting as the basis for comparing volunteer decision approaches.

2.1 Game-theoretic volunteer coordination

In the baseline process, pedestrians observe nearby obstacles and visible agents during a volunteer selection step. A volunteer decision model estimates each pedestrian's willingness to help. Potential volunteers are grouped by the obstacles they can assist, and the nearest candidate is selected for each obstacle. Other pedestrians continue evacuating. Selected volunteers move toward their assigned obstacles and remove them, restoring access to affected routes. Between selection steps, current assignments are preserved.

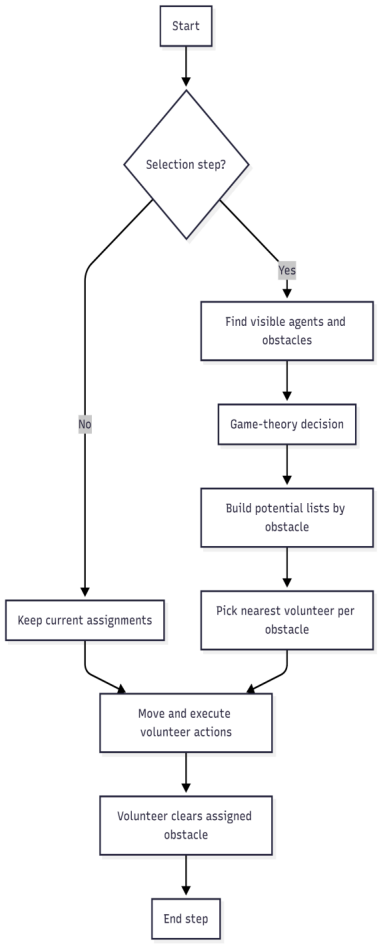


Figure 1. Baseline volunteer coordination and obstacle-removal workflow.

Figure 1 shows the sequence used for the baseline comparison. The distinction between selection steps and action execution separates the decision to volunteer from the movement and obstacle-removal stages shared by the two approaches.

2.2 Volunteer probability model

Figure 2 presents the baseline probability model. Here, N_{near} is the number of nearby pedestrians, b is the obstacle blockage ratio, ω is an unwillingness coefficient, C_v is the volunteer cost, and C_f is the failure cost. The cost ratio β is clamped to $[10^{-4}, 1]$, and $x = \max(2, N_{near})$. A uniform random sample determines whether a pedestrian becomes a potential volunteer.

$$P_{vol} = 1 - \left(1 - e^{-\omega b}\right)^{1/x} \cdot \beta^{1/(x-1)}$$

$$\beta = \text{clamp}\left(\frac{C_v}{C_f}, 10^{-4}, 1\right)$$

$$x = \max(2, N_{near})$$

$$Volunteer = \mathbf{1}\left[U(0,1) < P_{vol}\right]$$

Figure 2. Game-theoretic volunteer probability and Bernoulli sampling rule.

For fixed x and nonnegative ω and b, the displayed equation makes the volunteer probability decrease as b increases. Increasing β also decreases the probability, holding the other variables fixed. These relationships follow the equation in Figure 2. Candidate selection is followed by the obstacle-specific assignment process described in Section 2.1.

3 Proposed LLM Volunteer Coordination

3.1 Context and decision inputs

The proposed framework uses evacuation context to inform volunteer decisions. At each selection step, visible obstacles and nearby pedestrians are identified, and candidate information is prepared for the LLM. Table 1 summarizes the inputs described in the framework.

Table 1. Context supplied for LLM-based volunteer decisions.

Input	Description
Visible obstacles	Obstacles within the observation range
Nearby pedestrians	Pedestrians around each obstacle
Available volunteers	Candidates available to assist
Distance	Distance between a candidate and an obstacle
Obstacle impact	Estimated influence of removal on evacuation

The intended coordination process considers candidate proximity and obstacle impact when assigning volunteers. Its natural-language output records contextual reasons associated with the decision, allowing volunteer behavior to be examined alongside evacuation performance.

3.2 Decision and execution workflow

Figure 3 preserves the proposed workflow. At a selection step, the framework checks whether the LLM is available and whether the request limit permits a call. The LLM branch produces volunteer intent and coordinated assignments. If a call is unavailable or the limit is reached, the workflow provides a mock or greedy fallback. Outside a selection step, existing assignments remain in effect.

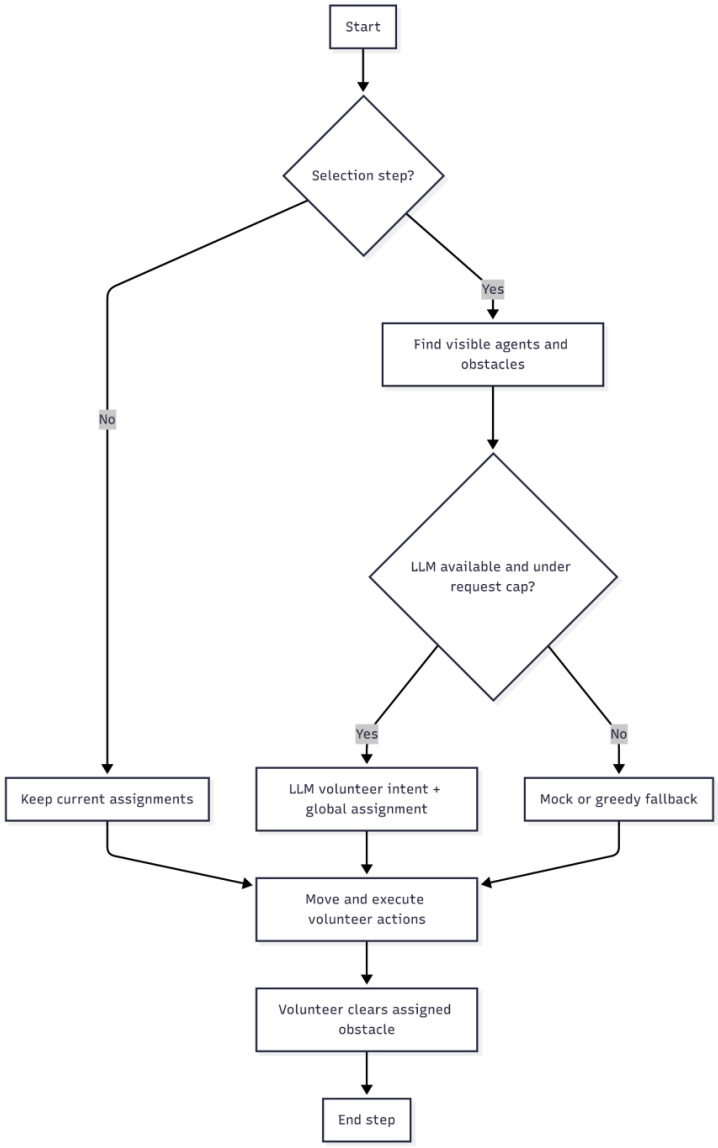


Figure 3. LLM volunteer decision workflow, including request checks and fallback handling.

All branches return to volunteer movement and action execution. Assigned volunteers approach and clear their obstacles before the step ends. The fallback branch distinguishes the intended LLM workflow from continuous dependence on remote inference; its use should be recorded when evaluating the decision policy.

3.3 Simulation architecture

The architecture consists of a simulation core, an environment layer, an agent layer, a decision and game layer, and a data export component. Figure 4 shows their interactions and retains the original architecture diagram.

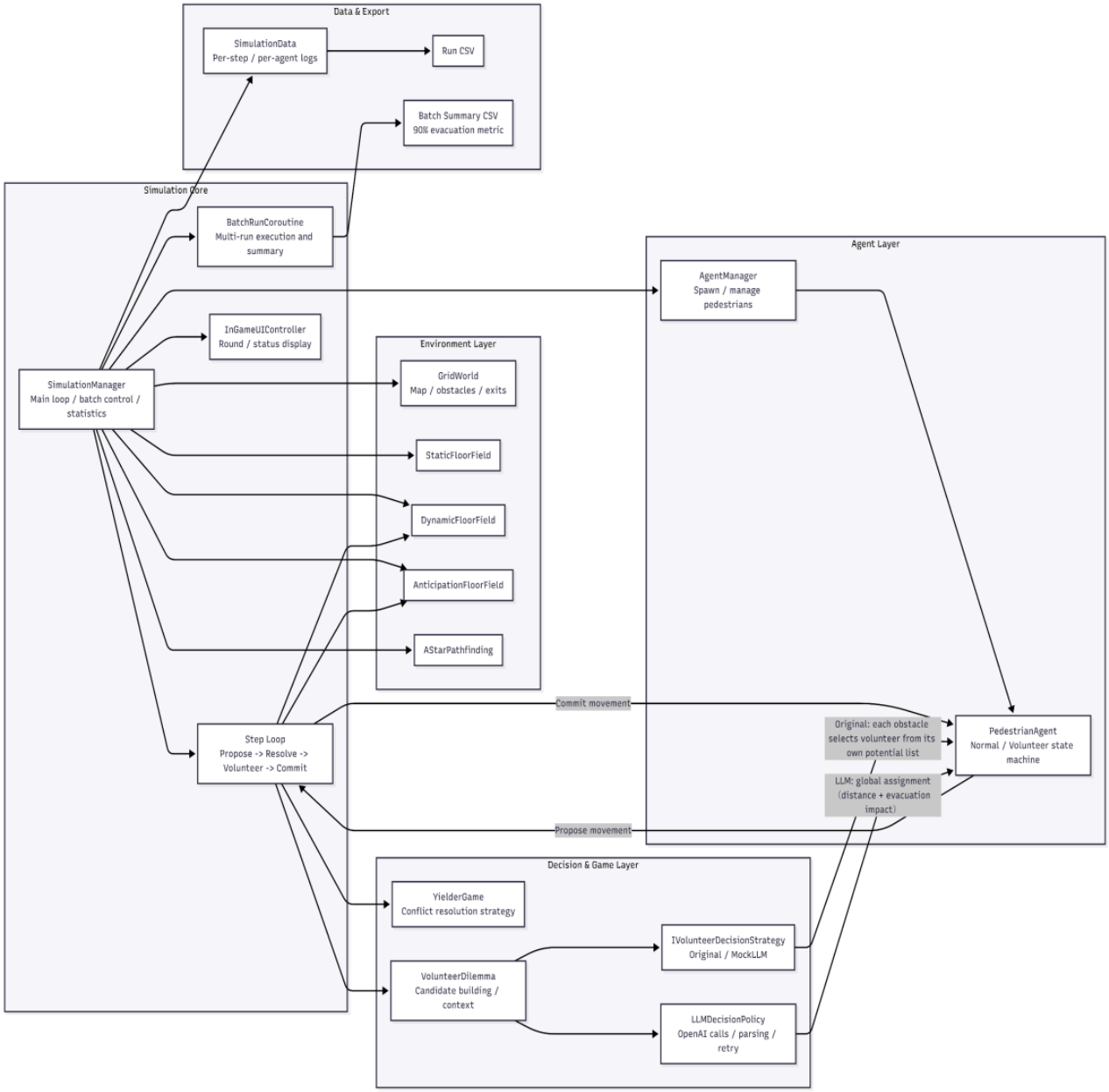
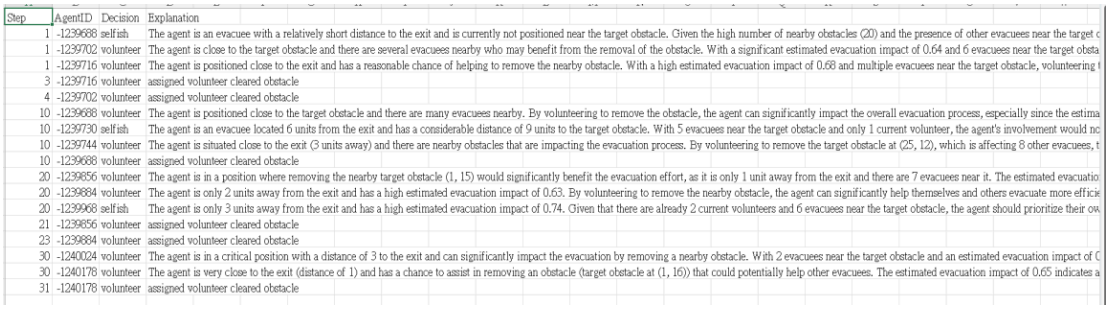


Figure 4. Simulation architecture and the integration of volunteer decision policies.

The simulation core controls the step loop and batch execution. Environment components maintain the grid and navigation fields, while the agent layer manages pedestrian states and movement. The decision layer supplies volunteer decisions and conflict resolution. Data export records run logs and summary metrics. The comparison changes the volunteer decision policy within this shared design.

3.4 Interpretable volunteer decisions

The LLM produces natural-language explanations in addition to volunteering decisions. Figure 5 shows an example decision log containing simulation steps, agent identifiers, decisions, and explanations. These outputs identify factors such as proximity to an obstacle, estimated evacuation impact, and nearby pedestrians.



Step	AgentID	Decision	Explanation
1	-1239688	selfish	The agent is an evacuee with a relatively short distance to the exit and is currently not positioned near the target obstacle. Given the high number of nearby obstacles (20) and the presence of other evacuees near the target c
1	-1239702	volunteer	The agent is close to the target obstacle and there are several evacuees nearby who may benefit from the removal of the obstacle. With a significant estimated evacuation impact of 0.64 and 6 evacuees near the target obsta
1	-1239716	volunteer	The agent is positioned close to the exit and has a reasonable chance of helping to remove the nearby obstacle. With a high estimated evacuation impact of 0.68 and multiple evacuees near the target obstacle, volunteering t
3	-1239716	volunteer	assigned volunteer cleared obstacle
4	-1239702	volunteer	assigned volunteer cleared obstacle
10	-1239688	volunteer	The agent is positioned close to the target obstacle and there are many evacuees nearby. By volunteering to remove the obstacle, the agent can significantly impact the overall evacuation process, especially since the estima
10	-1239730	selfish	The agent is an evacuee located 6 units from the exit and has a considerable distance of 9 units to the target obstacle. With 5 evacuees near the target obstacle and only 1 current volunteer, the agent's involvement would nc
10	-1239744	volunteer	The agent is situated close to the exit (3 units away) and there are nearby obstacles that are impacting the evacuation process. By volunteering to remove the target obstacle at (25, 12), which is affecting 8 other evacuees, t
10	-1239688	volunteer	assigned volunteer cleared obstacle
20	-1239856	volunteer	The agent is in a position where removing the nearby target obstacle (1, 15) would significantly benefit the evacuation effort, as it is only 1 unit away from the exit and there are 7 evacuees near it. The estimated evacuatio
20	-1239884	volunteer	The agent is only 2 units away from the exit and has a high estimated evacuation impact of 0.63. By volunteering to remove the nearby obstacle, the agent can significantly help themselves and others evacuate more efficie
20	-1239968	selfish	The agent is only 3 units away from the exit and has a high estimated evacuation impact of 0.74. Given that there are already 2 current volunteers and 6 evacuees near the target obstacle, the agent should prioritize their ow
21	-1239856	volunteer	assigned volunteer cleared obstacle
23	-1239884	volunteer	assigned volunteer cleared obstacle
30	-1240024	volunteer	The agent is in a critical position with a distance of 3 to the exit and can significantly impact the evacuation by removing a nearby obstacle. With 2 evacuees near the target obstacle and an estimated evacuation impact of 0
30	-1240178	volunteer	The agent is very close to the exit (distance of 1) and has a chance to assist in removing an obstacle (target obstacle at (1, 16)) that could potentially help other evacuees. The estimated evacuation impact of 0.65 indicates a
31	-1240178	volunteer	assigned volunteer cleared obstacle

Figure 5. Example of LLM volunteer decisions and explanations from the simulation log.

For example, the log includes the explanation “The agent is close to the target obstacle” before discussing nearby evacuees and estimated impact. Explanations provide material for qualitative review, but their presence does not establish that they faithfully describe the model’s internal reasoning or independently improve performance.

4 Experimental Setup

4.1 Simulation conditions

Table 2. Experimental conditions.

Parameter	Setting
Environment	Grid-based evacuation simulation
Number of exits	4
Obstacle placement	Randomly generated
Obstacle counts evaluated	20, 40, 60, 80
Agent populations evaluated	60, 90, 120, 150
Repeated runs	30 per scenario
Volunteer selection interval	Every 6 simulation steps

The evaluation compares the game-theoretic baseline and the LLM-based approach under the conditions in Table 2. Repetition addresses stochastic variation in the simulation. The results plots summarize performance by agent population; the mapping or aggregation of obstacle-count conditions is not specified, so separate effects of obstacle count cannot be inferred from these plots.

4.2 Evaluation measures

The primary measure is T90, the number of simulation steps required for at least 90% of the initial population to evacuate. Lower T90 indicates faster evacuation in simulation steps. Relative improvement is defined as $100 \times (T90_{baseline} - T90_{LLM}) / T90_{baseline}$. This measure does not include remote inference latency or monetary cost.

5 Results

5.1 Evacuation performance

Figure 6 compares the reported T90 values at the four agent populations. The LLM-based curve lies below the baseline curve at each population. Both curves increase with population, and the difference between them is larger at the higher populations in the evaluated setting.

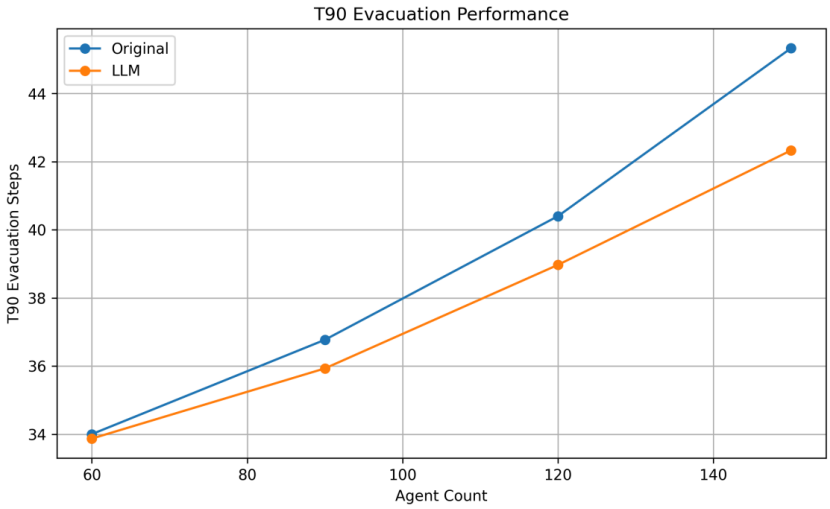


Figure 6. T90 evacuation performance of the baseline and LLM-based approaches.

5.2 Relative improvement

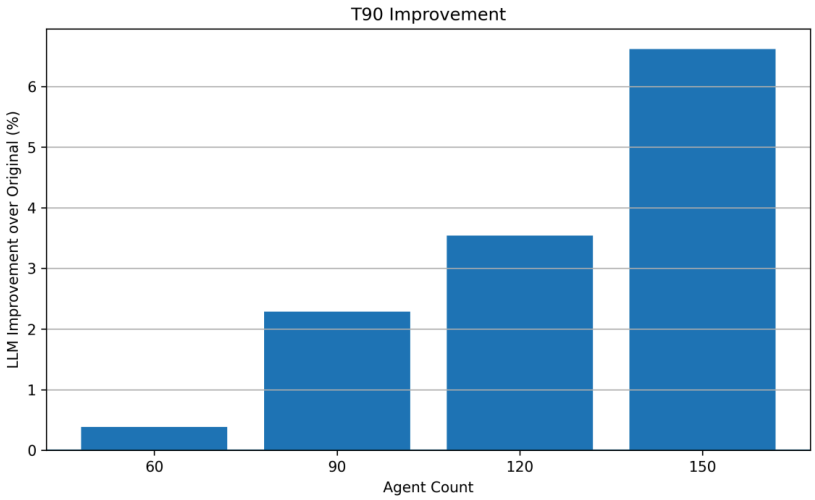


Figure 7. Reported relative T90 improvement across the evaluated agent populations.

Figure 7 shows the same increasing trend in relative improvement as population rises. Context-sensitive volunteering is a possible explanation for this pattern, but the plots do not isolate the causal contribution of coordination, prompting, or individual context variables. The qualitative log in Figure 5 complements these results by showing the factors mentioned in individual decisions.

The results are descriptive. The plots do not include uncertainty intervals or significance tests, and the underlying per-run measurements are not presented here. Consequently, the observed differences should not be interpreted as statistically established improvements or as evidence of performance beyond the evaluated conditions.

6 Discussion

6.1 Potential benefits

The evaluated framework suggests that contextual reasoning may support volunteer decisions in evacuation environments with obstacles. Considering distance, pedestrian density, and estimated obstacle impact gives the decision policy information about both the feasibility and potential benefit of volunteering. The reported trend at larger agent populations motivates further study of how these factors influence evacuation efficiency.

The proposed coordination design aims to consider multiple candidates and obstacles when forming assignments. However, a lower T90 alone does not demonstrate that global coordination is responsible for the result. An ablation study comparing local decisions, coordinated assignments, and different context inputs would help determine which components contribute to performance.

Natural-language explanations offer a readable record for reviewing volunteer choices. They can help identify unexpected decisions and support analysis of the supplied context. Their usefulness should be assessed separately from numerical evacuation performance, including whether the explanations are consistent with the recorded state and final action.

6.2 Limitations and reproducibility

Remote LLM calls incur API cost and inference latency. Because T90 is measured in simulation steps, faster simulated evacuation does not necessarily imply lower wall-clock execution time. The quality of decisions may also depend on the selected model and prompt. Model identity, prompt content, inference settings, request limits, and fallback usage should therefore accompany a reproducible evaluation.

The present comparison is limited to a grid-based environment and the evaluated populations. Random obstacle placement introduces variation, and the available figures do not report per-run distributions, matched random seeds, or uncertainty estimates. Future experiments should document these conditions, clarify how obstacle counts are paired or aggregated, and distinguish remote LLM decisions from fallback decisions.

7 Conclusion and Future Work

This study examined LLM-based volunteer coordination within a crowd evacuation simulation and compared it with a game-theoretic baseline. The reported results show lower T90 values for the LLM-based approach across the evaluated agent populations, with larger differences at higher populations. Natural-language explanations additionally support qualitative inspection of volunteer behavior. These observations motivate further evaluation of contextual decision policies under controlled and reproducible conditions.

Future work should evaluate more complex environments, including dynamic obstacles and changing evacuation conditions. Comparisons among different LLMs and locally deployed models could examine inference cost and response time. Statistical analysis of repeated runs, ablation experiments, and validation using human behavioral or real-world evacuation data would help assess robustness and practical relevance.

References

[1] G. W. Lin and S. K. Wong, “Evacuation simulation with consideration of obstacle removal and using game theory,” *Physical Review E*, vol. 97, no. 6, article 062303, 2018. doi:10.1103/PhysRevE.97.062303.